

26. What is a union?

Answer: A union allows different members to share the same memory location.

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27. What is a header file?

Answer: A header file contains function declarations and macros.

Example:

```
#include <stdio.h>
```

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28. What is dynamic memory allocation?

Answer: It allocates memory during program execution using functions like `malloc()`, `calloc()`, `realloc()`, and `free()`.

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## 29. What is the difference between compile-time and run-time errors?

Answer:

- Compile-time errors occur during compilation.
- Run-time errors occur while the program is running.

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30. Why is C called a middle-level language?

Answer: Because it supports both low-level programming (memory access using pointers) and high-level programming (functions, loops, structures).

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